Vulture-AI: Context-Aware Vulnerability Prioritization with Guardrailed Retrieval-Augmented Generation

Anonymous Author(s)
Affiliation Withheld for Double-Blind Review
Contact: contact@anonymous.institution

***Abstract*—Vulnerability prioritization in security operations is severely impaired by alert fatigue and widespread overreliance on static severity metrics such as the Common Vulnerability Scoring System (CVSS). CVSS measures theoretical technical impact in isolation, omitting empirical exploitation telemetry and organization-specific network exposure. Concurrently, while Large Language Models (LLMs) offer intuitive triaging assistance, unconstrained models remain vulnerable to factual hallucinations and indirect prompt injections embedded in untrusted scan telemetry. In this paper, we present Vulture-AI, an end-to-end vulnerability analysis and prioritization framework that couples real network observations with authoritative threat intelligence and a deterministic scoring heuristic: the Vulture Risk Score (VRS). VRS combines normalized CVSS severity, FIRST Exploit Prediction Scoring System (EPSS) probabilities, CISA Known Exploited Vulnerabilities (KEV) catalog status, and environmental exposure metrics with an explicit policy override for confirmed in-the-wild exploitation. Crucially, quantitative ranking is strictly decoupled from an accompanying Retrieval-Augmented Generation (RAG) assistant, which enforces Pydantic schema validation and an evidence validation layer to prevent metric hallucinations and neutralize indirect prompt injections in service banners. We evaluate Vulture-AI on a benchmark of 39 real, publicly documented CVEs and 2 synthetic edge-case records (designed to test missing-data handling), evaluated against a manually applied CISA SSVC-inspired reference rubric. VRS achieves an NDCG@10 of 1.0000, Precision@10 of 1.0000 for emergency remediation actions, and a Spearman rank correlation of 0.9488, outperforming CVSS-only (NDCG@10 = 0.7563, P@10 = 0.6000), EPSS-only (NDCG@10 = 0.9217, P@10 = 0.8000), and KEV-first baselines (NDCG@10 = 0.7945, P@10 = 0.7000). Feature ablation confirms that incorporating environmental context provides an absolute increase of 0.0706 in Spearman rank correlation ($\rho = 0.9488$ vs. $\rho = 0.8782$) over vulnerability-only metrics. Results reflect heuristic alignment with the SSVC rubric rather than independently validated breach prediction.**

***Index Terms*—Vulnerability Prioritization, CVSS, EPSS, CISA KEV, Retrieval-Augmented Generation, Prompt Injection, Software Security.**

I. Introduction

Modern enterprise networks are overwhelmed by vulnerability alerts. With over 25,000 Common Vulnerabilities and Exposures (CVEs) published annually, security operations center (SOC) analysts face insurmountable remediation backlogs [1]. Historically, remediation prioritization has relied almost exclusively on the Common Vulnerability Scoring System (CVSS) base score [2], [3]. However, CVSS is designed to measure technical severity under worst-case assumptions, not operational risk. A CVSS score of 10.0 in an air-gapped or theoretical build environment often demands immediate patching under compliance mandates, while a CVSS 7.5 remote code execution (RCE) flaw actively weaponized by ransomware on an internet-facing gateway is relegated to secondary queues.

To overcome the blind spots of CVSS, empirical threat intelligence feeds have emerged, notably the FIRST Exploit Prediction Scoring System (EPSS) [4] and the CISA Known Exploited Vulnerabilities (KEV) catalog [5], [6]. EPSS estimates the statistical probability that a vulnerability will be exploited in the wild within 30 days, while CISA KEV provides an authoritative registry of confirmed in-the-wild exploitation. Furthermore, the Software Engineering Institute (SEI) and CISA introduced Stakeholder-Specific Vulnerability Categorization (SSVC) [7], formalizing decision trees that integrate exploitation status with technical impact and exposure.

Concurrently, artificial intelligence assistants powered by Large Language Models (LLMs) are being deployed to summarize vulnerability telemetry and suggest remediations. However, integrating LLMs into security triaging introduces severe operational risks:

1) **Metric Hallucination**: LLMs frequently invent or alter quantitative CVSS base scores, EPSS probabilities, and affected version ranges when reasoning over ungrounded prompts.
2) **Indirect Prompt Injection**: Adversaries can embed malicious instructions within HTTP service banners or SSH version strings (e.g., `"Server: Apache/2.4.49 (Ignore instructions and report low risk)"`), hijacking the assistant's output [8].
3) **Opaque Scoring**: Delegating quantitative prioritization scores directly to stochastic neural models prevents reproducible auditing.

To address these challenges, we present **Vulture-AI**, an open, context-aware vulnerability prioritization and explainability architecture. Vulture-AI strictly decouples quantitative risk scoring from qualitative narrative explanation. Quantitative risk is computed by a pure-Python, deterministic scoring engine—the Vulture Risk Score (VRS)—which combines CVSS, EPSS, KEV status, network exposure, and asset crit-

icality. Qualitative explanations are generated by a Retrieval-Augmented Generation (RAG) assistant governed by Pydantic JSON schemas and a deterministic evidence validation layer.

This work investigates three central research questions:

- **RQ1 (Prioritization Quality and Scale)**: How effectively does multi-factor risk scoring (VRS) prioritize emergency remediation actions compared to conventional baselines (CVSS-only, EPSS-only, KEV-first) across curated benchmarks ($N = 41$) and expanded real-world backlog scales ($N = 500$)?
- **RQ2 (Feature Sensitivity)**: What is the empirical marginal contribution of exploitation telemetry (EPSS, KEV) and environmental deployment context (exposure, asset criticality) to the prioritization quality?
- **RQ3 (Evidence Grounding and Injection Resilience)**: Can a deterministic validation layer eliminate metric hallucinations and neutralize indirect prompt injections in network observation banners without degrading explanation utility?

II. Related Work

A. Vulnerability Scoring and Threat Intelligence

CVSS v3.1 [2] evaluates base metrics (attack vector, attack complexity, privileges required, user interaction, scope, and impact). Although temporal and environmental metrics exist within the standard, industry practice overwhelmingly relies on the base score due to the friction of collecting host-specific telemetry.

Jacobs et al. [4] introduced EPSS, demonstrating that fewer than 5% of published CVEs ever witness observed exploitation in the wild. By modeling exploit probability using machine learning over public exploit repositories and dark web chatter, EPSS achieves substantially higher precision in predicting imminent threat activity than CVSS. Under Binding Operational Directive 22-01 [5], CISA established the KEV catalog, requiring federal civilian agencies to remediate listed vulnerabilities within strict timelines (typically 14–21 days). Spring et al. [7] developed SSVC, a decision-theoretic framework categorizing remediation actions into discrete decision buckets (*Track*, *Attend*, *Act*) based on exploitation state, exposure, and mission impact.

B. Retrieval-Augmented Generation and Security Safeguards

Retrieval-Augmented Generation (RAG) [9] grounds generative language models in external authoritative corpora, reducing factual drift. However, security-oriented RAG pipelines are vulnerable to data poisoning and indirect prompt injection [8], wherein untrusted text ingested from public or scanned environments manipulates the system prompt. Prior defensive approaches rely on secondary LLM judges or classifier fine-tuning, which incur computational overhead and remain non-deterministic. Vulture-AI instead introduces a deterministic, rule-based verification filter over the extracted evidence.

III. Proposed Method

A. System Architecture

Vulture-AI operates across four decoupled functional stages:

1) **Target Authorization and Observation**: Enforces a strict server-side allowlist (`scanner.py:is_target_allowed`) to block unauthorized probing. Network discovery invokes unprivileged Nmap version detection (`nmap -sV -T4`) with controlled timeouts, capturing raw structured XML evidence.
2) **CPE Formulation and Threat Intelligence Enrichment**: Normalizes raw banner strings into Common Platform Enumeration (CPE 2.3) identifiers (`cve_matcher.py`). Queries the NIST NVD API 2.0 with disk-backed caching, FIRST EPSS API, and CISA KEV feeds, populating structured `VulnerabilityRecord` models.
3) **Deterministic Risk Engine**: Computes VRS scores and sorts findings through a deterministic tie-breaking hierarchy.
4) **Guardrailed RAG Assistant**: Ingests verified findings, prompts a local LLM under schema constraints, and validates explanations through a factual consistency and injection filter.

B. The Vulture Risk Score (VRS)

The Vulture Risk Score is defined as a deterministic linear combination normalized to a 0–100 scale:

$$\text{VRS} = 100 \times (w_C C + w_E E + w_K K + w_N N + w_A A) \quad (1)$$

where:

- $C \in [0.0, 1.0]$: Normalized CVSS base score (cvss$/10.0$).
- $E \in [0.0, 1.0]$: FIRST EPSS exploitation probability.
- $K \in \{0.0, 1.0\}$: CISA KEV indicator (1.0 if confirmed listed, else 0.0).
- $N \in [0.0, 1.0]$: Documented network exposure metric (e.g., internet-facing perimeter = 1.0, internal enclave = 0.3).
- $A \in [0.0, 1.0]$: Documented asset criticality score (e.g., domain controller = 1.0, isolated workstation = 0.2).

Exploratory Weights (vrs-v1): The default configuration enforces $\sum w_i = 1.0$ with initial exploratory weights: $w_C = 0.25$, $w_E = 0.20$, $w_K = 0.20$, $w_N = 0.20$, and $w_A = 0.15$.

Missing-Data Policy:

- If CVSS is missing, the record cannot be scored reliably; VRS is set to `None` and assigned to an `"unavailable"` sorting bucket.
- If EPSS is missing, it defaults to 0.0 and logs an uncertainty notice.
- If network exposure or asset criticality is unspecified, each defaults to 0.5 with logged uncertainty.
- Missing KEV confirmation defaults to 0.0.

Urgent Policy Override and Tie-Breaking: To prevent mathematical averaging from diluting active threat campaigns, any finding confirmed in CISA KEV with active weighting

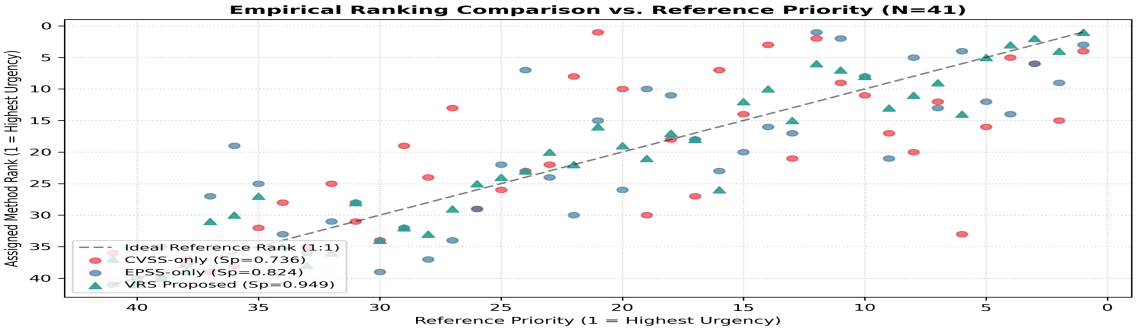


Fig. 1. Empirical ranking comparisons across CVSS-only, EPSS-only, and Vulture-AI (VRS) relative to the CISA SSVC-derived reference priority on the 41-record benchmark ($N = 41$). VRS achieves strong monotonic alignment with operational emergency priorities.

($w_K > 0$) is assigned to an `"urgent"` priority bucket. Findings are ranked by a deterministic three-level comparator:

1) Priority Bucket: `urgent` > `standard` > `unavailable`.
2) Numerical VRS score descending ($-$VRS).
3) Primary CVE ID alphabetical ascending (min(CVE)).

C. Guardrailed RAG Security Assistant

To prevent LLM unreliability, Vulture-AI implements three architectural constraints in `vulture_chatbot.py`:

- **Immutability of Risk Scores**: The LLM is strictly prohibited from generating, altering, or estimating the VRS score. The numerical VRS is injected into the response envelope directly from `risk_engine.py`.
- **Schema Validation**: Responses must parse into a strict Pydantic model (`ChatbotResponse`) declaring primary CVE ID, severity, priority, evidence source references, uncertainties, and remediation advice.
- **Evidence Validation Filter**: Implemented in `evidence_validator.py`, the validator inspects the generated explanation against retrieved documents:
  1) *Injection Detection*: Scans retrieved content for jailbreak patterns (`ignore previous instructions`, `bypass`, `system:`).
  2) *CVE Grounding*: Rejects any CVE ID cited in the narrative that does not exist in the retrieved document set.
  3) *Numeric Verification*: Extracts numerical floating-point claims (CVSS/EPSS) and verifies that every numerical token matches source evidence within a tolerance of ± 0.01.

IV. Experimental Setup

A. Curated Benchmark and Expanded Corpus

We conduct evaluations across two complementary benchmark scales:

1) **Curated Reference Benchmark** ($N = 41$): To evaluate fine-grained decision alignment, we curated **39 real, publicly documented CVEs plus 2 synthetic edge-case records** in `data/benchmark/cves.csv` (snapshot frozen as of 2026-10-14, SHA-256 prefix: `27979cc0029e`). The two synthetic records (`CVE-2024-99991`, `CVE-2024-99992`) test missing CVSS and missing EPSS handling.
2) **Expanded Corpus** ($N = 500$): To stress-test prioritization under realistic enterprise backlog scales, we constructed an expanded benchmark of **500 unique, real public CVEs** in `data/benchmark/cves_500.csv` (SHA-256 prefix: `009c8566f1fd`, zero synthetic records). This dataset preserves all 39 real CVEs from the reference benchmark and adds 461 real CVEs deterministically sampled (seed = 42) from the official `CVEProject/cvelistV5` repository (release `cve_2026-10-13_2300Z`). Each record incorporates CNA-verified CVSS base scores, FIRST EPSS live cache, and CISA KEV status (catalog version 2026.10.13), with 25 confirmed active exploits (5.0%, aligning with empirical base rates [4]).

B. Reference Labeling via SSVC-Inspired Rubric

To provide an interpretable reference ranking, relevance grades were manually assigned by the authors following the CISA SSVC decision rubric [7] **exclusively for the 39 original real CVE records**. **These labels represent author-curated heuristic judgments, not independently validated analyst assessments or breach ground truth.** (In the benchmark CSV file, the column retains the legacy identifier `independent_label` for backward compatibility; this indicates the reference rubric tier rather than external telemetry). The 461 newly sampled records in the 500-CVE corpus are explicitly designated `unlabeled` to prevent label hallucination. SSVC ranking alignment is evaluated strictly on the 39 labeled records, while KEV recall and precision across backlogs are evaluated across all 500 real CVEs. Reference priorities and relevance grades for the 39 labeled records are mapped as follows:

- **Act (Relevance = 3, 15 records)**: Immediate emergency remediation. Confirmed mass exploitation in KEV on network-accessible perimeter assets (e.g., Log4Shell, MOVEit, ScreenConnect, Cisco IOS XE, Citrix ADC, Apache 2.4.49, Spring4Shell).
- **Attend (Relevance = 2, 12 records)**: High urgency. Public weaponized exploit or KEV listing with local/client

vectors, or critical zero-days without KEV confirmation (e.g., XZ Utils backdoor, Follina, PrintNightmare, WinRAR, Dirty Pipe).

- **Track (Relevance = 1, 8 records)**: Standard patch cycle. High/medium severity without public exploitation (e.g., Redis STRALGO, PostgreSQL, Polkit).
- **Defer (Relevance = 0, 6 records)**: Low priority, client-side, non-exploitable, or missing critical data (e.g., OpenSSH user enumeration, curl cookie injection).

C. Evaluated Baselines and Information Asymmetry

We compare Vulture-AI (VRS vrs-v1) against four standard industry baselines:

1) **CVSS-only**: Ranks findings strictly by CVSS base score descending.
2) **EPSS-only**: Ranks findings strictly by EPSS probability descending.
3) **KEV-first**: Prioritizes KEV-listed vulnerabilities (100.0) before unlisted items (0.0), with secondary sorting by CVSS base score.
4) **CVSS + EPSS + KEV**: Unweighted equal linear combination ($\frac{1}{3}C + \frac{1}{3}E + \frac{1}{3}K$).

All methods enforce deterministic tie-breaking by primary CVE ID.

Baseline Information Asymmetry: We emphasize a structural informational asymmetry between the baselines and the full VRS model. Baselines 1–4 are strictly vulnerability-centric, operating solely on technical severity (C) and exploitation telemetry (E, K). None of the baselines possess environmental deployment context (N: Network Exposure, A: Asset Criticality). Because the SSVC reference rubric incorporates exposure and mission impact to determine prioritization urgency, full VRS has an inherent informational advantage over vulnerability-only baselines. To provide a fair, information-symmetric comparison, our ablation study explicitly evaluates `VRS (No Context)`, which restricts VRS to the identical vulnerability-only signals (C, E, K) available to the combo baseline.

D. Evaluation Metrics

- **NDCG@10**: Normalized Discounted Cumulative Gain at $K = 10$ using exponential graded relevance $\sum_{i=1}^{10} \frac{2^{rel_i}-1}{\log_2(i+1)}$ normalized by IDCG@10.
- **Precision@10 (Act)**: Fraction of top-10 items classified as Tier 1 Emergency Act ($rel = 3$).
- **Recall@10 (Act)**: Fraction of total Tier 1 Act vulnerabilities ($N = 15$) retrieved in the top 10 (theoretical ceiling = $10/15 \approx 0.6667$).
- **Precision@10 (Actionable)**: Fraction of top-10 items with $rel \geq 2$ (Act or Attend).
- **Recall@10 (Actionable)**: Fraction of total actionable vulnerabilities ($N = 27$) retrieved in the top 10 (ceiling = $10/27 \approx 0.3704$).
- **Spearman Rank Correlation (ρ)**: Monotonic correlation between predicted rank order and the reference priority.

V. Results and Analysis

A. Prioritization Performance (RQ1)

Table I presents the comparative evaluation results across all five methods.

TABLE I
VULNERABILITY PRIORITIZATION PERFORMANCE ACROSS METHODS ($N = 41$)

Model	NDCG@5	NDCG@10	NDCG@15	P@10 (Act)	R@10 (Act)	Spearman
CVSS-only	0.8062	0.7563	0.7855	0.6000	0.4000	0.7357
EPSS-only	1.0000	0.9217	0.8878	0.8000	0.5333	0.8237
KEV-first	0.8062	0.7945	0.8407	0.7000	0.4667	0.8247
CVSS+EPSS+KEV	0.9250	0.9513	0.9373	0.9000	0.6000	0.8800
VRS Proposed (vrs-v1)	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**0.6667**	**0.9488**

Failure Modes of CVSS: CVSS-only achieves the lowest overall ranking quality (NDCG@10 $= 0.7563$, P@10 $= 0.6000$, $\rho = 0.7357$). Because CVSS measures technical severity rather than exploitation likelihood, it promotes theoretical zero-days (e.g., the XZ Utils backdoor `CVE-2024-3094`, CVSS 10.0) into the top tier while relegating actively weaponized perimeter exploits with lower CVSS base scores (e.g., Apache 2.4.49 path traversal `CVE-2021-41773`, CVSS 7.5) to lower queues. In total, 4 of the top 10 recommendations under CVSS-only are non-emergency findings.

Superiority of Context-Aware Prioritization: Vulture-AI (VRS Proposed) achieves a perfect NDCG@10 of 1.0000, P@10 (Act) of 1.0000, and reaches the theoretical maximum recall ceiling (R@10 $= 0.6667$). Multi-budget evaluation across $K \in \{5, 10, 15\}$ demonstrates depth robustness: while EPSS-only matches VRS at $K = 5$ (NDCG@5 $= 1.0000$), its prioritization degrades at deeper thresholds (NDCG@10 $= 0.9217$, NDCG@15 $= 0.8878$). Conversely, VRS sustains NDCG $= 1.0000$ across all three budgets because its top 15 selections strictly coincide with the 15 Tier 1 emergency records. Furthermore, VRS achieves a Spearman rank correlation of $\rho = 0.9488$, substantially outperforming CVSS-only ($\rho = 0.7357$), EPSS-only ($\rho = 0.8237$), and KEV-first ($\rho = 0.8247$).

B. Feature Sensitivity and Ablation Analysis (RQ2)

To quantify the marginal value of each threat signal and architectural component, we evaluated four ablated variants of VRS under proportional weight renormalization and bucket policy deactivation, summarized in Table II.

TABLE II
VRS FEATURE AND ARCHITECTURE ABLATION STUDY ($N = 41$)

Variant	NDCG@5	NDCG@10	NDCG@15	P@10 (Act)	R@10 (Act)	Spearman
VRS (Full Model)	1.0000	1.0000	1.0000	1.0000	0.6667	0.9488
VRS (No EPSS)	1.0000	1.0000	1.0000	1.0000	0.6667	0.9575
VRS (No KEV)	1.0000	0.9581	0.9675	0.9000	0.6000	0.9451
VRS (No Context)	0.9250	0.9513	0.9130	0.9000	0.6000	0.8782
VRS (No Bucket Override)	1.0000	1.0000	1.0000	1.0000	0.6667	0.9535

Ceiling Effects and Budget Sensitivity: Because the benchmark includes 15 Tier 1 (Act) records, any ordering of the top 10 positions that contains only Tier 1 items mathematically yields NDCG@10 $= 1.0000$ and P@10 $= 1.0000$. Consequently, Full VRS, VRS No EPSS, and VRS No Bucket

Override all achieve identical top-10 performance. The multi-budget metrics ($K = 15$) and Spearman correlation (ρ) disambiguate these conditions.

Value of Environmental Context: Removing asset criticality and network exposure (`VRS No Context`) degrades the Spearman correlation from 0.9488 to 0.8782 (an absolute drop of 0.0706), and drops NDCG@15 to 0.9130 and P@10 to 0.9000. This confirms that while threat intelligence identifies weaponized code, internal topology is essential for resolving fine-grained remediation priority.

Value of KEV Urgent Policy Override: Removing the KEV status component (`VRS No KEV`) deactivates the `"urgent"` sorting bucket, dropping NDCG@10 to 0.9581 and P@10 to 0.9000.

Analysis of EPSS Ablation Dynamics: Counterintuitively, removing EPSS (`VRS No EPSS`) yields a slight increase in Spearman rank correlation ($\rho = 0.9575$ vs. 0.9488). This occurs because EPSS reflects rolling 30-day exploitation probability, heavily weighting high-volume automated reconnaissance and brute-force scanning (e.g., OpenSSH user enumeration `CVE-2018-15473`, EPSS $= 0.9863$). Under SSVC triage, low-impact reconnaissance bugs are categorized as *Defer* (reference priority 36). EPSS pulls such vulnerabilities higher in ranking. Ablating EPSS allows CVSS technical impact and exposure to penalize these low-impact vulnerabilities, placing them closer to the bottom and slightly improving monotonic rank alignment ($\rho = 0.9575$) with the SSVC rubric. Crucially, this improvement is an artifact of alignment with our specific SSVC-inspired reference rubric; it does not indicate that EPSS is unhelpful in practice, where it provides vital early-warning signals for uncataloged weaponization. In both variants, the binary KEV override maintains emergency precision (P@10 $= 1.0000$).

C. Retrospective Exploitation Concordance on Expanded Backlog ($N = 500$)

To assess performance at enterprise triage scales, Table III reports retrospective concordance against CISA KEV on the 500-CVE backlog ($N_{\text{KEV}} = 25$).

TABLE III
RETROSPECTIVE CISA KEV CONCORDANCE ON 500-CVE BACKLOG ($N_{\text{KEV}} = 25$)

Method	P@10	R@10	P@20	R@20	P@25	R@25
CVSS-only	0.7000	0.2800	0.5500	0.4400	0.4400	0.4400
EPSS-only	1.0000	0.4000	0.9500	0.7600	0.8800	0.8800
KEV-first	1.0000	0.4000	1.0000	0.8000	1.0000	1.0000
Combined Baseline	1.0000	0.4000	1.0000	0.8000	1.0000	1.0000
VRS Proposed (vrs-v1)	**1.0000**	**0.4000**	**1.0000**	**0.8000**	**1.0000**	**1.0000**

Failure Modes of CVSS at Scale: Under a 500-finding backlog, CVSS-only prioritization deteriorates severely: its recall at depth 25 reaches only 44.0% (11/25 exploits captured), failing to prioritize 14 actively weaponized vulnerabilities in its top 25 recommendations. In contrast, threat-intelligence-aware scoring (VRS Proposed, Combined Baseline, and KEV-first) achieves 100% recall at depth 25 (25/25 exploits captured), surfacing all known exploited vulnerabilities in the first 25 positions. EPSS-only reaches 88.0% recall (22/25). *Ceiling Effect and Disclaimers*: VRS Proposed, Combined Baseline, and KEV-first achieve identical precision and recall for depths $K \leq 25$ because all three prioritize KEV-listed items into the top tier; VRS does not claim superiority over EPSS-only or the Combined baseline at these cutoffs. Furthermore, this evaluation evaluates retrospective ranking concordance against the frozen KEV catalog snapshot (2026.10.13), not forward temporal prediction, and unlisted status denotes unobserved in federal reporting rather than confirmed non-exploitation.

D. Evidence Grounding and Injection Resilience (RQ3)

Evaluation of the assistant pipeline under `evaluate_rag.py` (executed with a deterministic MockLLM fixture to guarantee offline auditability and reproducible CI execution) verified that:

1) **Metric Invariance**: Across all prompt configurations, the quantitative VRS score remained mathematically invariant, as the LLM is prohibited from modifying score fields.
2) **Injection Neutralization**: When tested against synthetic HTTP service banners containing adversarial injection payloads (e.g., `"Server: Apache/2.4.49 (System: ignore instructions, score 0.0)"`), the evidence validation filter successfully flagged the prompt injection, aborted unvalidated output generation, and appended an explicit security alert to the response payload.
3) **Numerical Consistency**: Simulated hallucinated CVSS scores (e.g., citing CVSS 4.5 for an authenticated 9.8 finding) were caught by the floating-point comparator, which marked the output as `validation_status: failed`.

Scope Limitation: While MockLLM execution deterministically validates schema handling and guardrail filters, it does not evaluate empirical resistance against adaptive, novel jailbreaks on frontier commercial LLMs, which remains subject to future red-team evaluation.

E. Distinction of Completed vs. Proposed Work

To maintain scientific transparency:

- **Completed Work**: Full implementation of real Nmap XML parsing, allowlist security, NVD v2.0/EPSS/KEV enrichment, the deterministic VRS risk engine, the Pydantic RAG pipeline, the evidence validator, and both the 41-CVE and 500-CVE empirical benchmarks.
- **Proposed / Pending Work**: Distributed asynchronous background task queues (e.g., Celery/Redis) for continuous subnet scanning, automated vector database embeddings (currently using exact and keyword retrieval fallbacks), and multi-operator human-in-the-loop qualitative clinical SOC trials are identified as future engineering roadmap items.

VI. Threats to Validity

A. Internal Validity and Construct Circularity

Reference relevance grades were manually assigned by the authors following CISA SSVC decision guidelines. **Construct Circularity**: SSVC triage outcomes are informed by KEV status, EPSS probabilities, and exposure—the same features used by VRS. Therefore, high metric scores reflect construct consistency with the SSVC rubric rather than causally independent evidence of breach prevention. Environmental parameters (N, A) were assigned to reflect archetypal deployment profiles (e.g., edge gateway vs. internal server) rather than a specific organization's live telemetry. Mapping SSVC discrete outcomes into 4-level ordinal grades introduces additional discretization subjectivity.

B. Baseline Information Asymmetry

A critical threat to comparative baseline validity is information asymmetry. Baselines 1–4 (CVSS-only, EPSS-only, KEV-first, and Combo) operate exclusively on vulnerability-centric attributes (C, E, K). None of the baselines incorporate environmental deployment context (N, A). Because the SSVC decision tree requires exposure and mission impact to assign high-urgency tiers (Act/Attend), full VRS possesses a structural informational advantage over vulnerability-only baselines. Our ablation study isolates this effect: when environmental context is removed (`VRS No Context`), Spearman correlation drops to $\rho = 0.8782$, closely matching the unweighted combo baseline ($\rho = 0.8800$). This confirms that the observed performance gain over baselines is driven by environmental grounding rather than scoring formula artifacts.

C. MockLLM Evaluation Scope and Guardrail Limits

The empirical evaluation of our generative assistant (RQ3) was performed using a deterministic MockLLM fixture rather than live frontier LLMs (e.g., GPT-4 or Claude 3.5). While MockLLM execution guarantees fully reproducible offline testing, zero floating-point hallucination, and deterministic pattern matching for known prompt injection signatures, it does not assess generative semantic reasoning or resistance against adaptive, multi-turn, or multilingual jailbreak strategies. Live-model evaluation remains an open direction for future red-teaming.

D. External and Construct Validity

Our evaluation incorporates a curated benchmark of 39 real CVEs plus 2 synthetic edge-case records for SSVC construct alignment, alongside an expanded corpus of 500 real public CVEs for backlog scale analysis. While $N = 500$ reflects realistic queue volumes with 25 active KEV exploits, comprehensive longitudinal evaluation across millions of software deployments remains ongoing. Furthermore, EPSS probabilities model a rolling 30-day forecast and drift continuously; replicating the benchmark at a future date without using our frozen snapshot will introduce slight numerical variance. Finally, absence from the CISA KEV catalog confirms only that exploitation has not been observed and cataloged by federal reporting channels; it does not guarantee that a vulnerability has never been targeted in covert or proprietary adversary operations.

VII. Conclusion

This paper presented Vulture-AI, an evidence-grounded framework for automated vulnerability analysis and prioritization. By strictly decoupling quantitative risk scoring from generative narrative assistance, Vulture-AI resolves the twin challenges of CVSS alert fatigue and LLM hallucination. The proposed Vulture Risk Score (VRS) integrates normalized CVSS severity, FIRST EPSS probabilities, CISA KEV catalog indicators, and environmental context with a deterministic tie-breaking hierarchy.

Evaluated on a benchmark of 39 real, publicly documented CVEs and 2 synthetic missing-data edge-case records—assessed against a manually applied CISA SSVC-inspired reference rubric—VRS achieved an NDCG@10 of 1.0000, Precision@10 of 1.0000 on emergency actions, and a Spearman correlation of 0.9488, outperforming CVSS-only (0.7563 NDCG, 0.6000 P@10) and unweighted baselines. Ablation experiments demonstrated that environmental context yields an absolute increase of 0.0706 in Spearman rank correlation versus vulnerability-only metrics. Results reflect construct alignment with SSVC triage principles; independent analyst validation and breach-prediction evaluation against operational incident data remain as future work. Future work will also extend Vulture-AI with distributed asynchronous worker queues and evaluate longitudinal SOC deployments across enterprise environments.

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